Michaela Mueller

Mark Slabodnick

Bio 210: Intro to Research

7 March 2025

# Effects of Temperature on Division in *Stentor coeruleus*

## Introduction

As climate change intensifies, understanding how temperature fluctuations affect various organisms is increasingly important. Temperature is a fundamental factor regulating biological processes such as metabolism, movement, and reproduction across all life forms (Schulte, 2015). Aquatic ecosystems, which support diverse life forms, are particularly vulnerable to rising temperatures, making them a critical focus for studying the impacts of climate change. While all organisms are affected by environmental shifts, smaller life forms, such as unicellular organisms, may be especially sensitive. In ciliates, temperature variations can profoundly influence cellular functions, division rates, and overall survival (Laybourn, 1975).

In this experiment, we examined the effects of temperature on *Stentor coeruleus*, a blue-pigmented ciliate known for its filter-feeding and regenerative abilities (Laybourn, 1975). Our previous studies explored how temperature influences *S. coeruleus* movement and regeneration. Due to their large size and visibility, stentors are commonly used in experimental research. We observed that *S. coeruleus* could not survive at temperatures above 35°C, as prolonged exposure led to cellular degradation. Conversely, at 10°C, stentors remained viable but exhibited significantly reduced movement and slower regeneration rates.

We designed this study to build on our previous observations by examining how temperature affects *S. coeruleus* reproduction. We selected 10°C, 20°C, and 30°C as our

experimental conditions, as these temperatures fall within the species' survivable range and may influence division rates. We hypothesized that lower temperatures would slow cell division due to metabolic suppression, while higher temperatures would promote higher division rates.

## Methods

### *Culture Conditions*

For this experiment, we used the single-celled organism *Stentor coeruleus*. The stentors were imported from North Carolina and held in a small container of spring water. They were kept at room temperature, approximately 22°C, and fed algae twice a week.

### *Experimental Setup*

We first set three incubators to the desired temperatures: 10°C, 20°C, and 30°C. Next, we put 25 mL of spring water in a large petri dish. We did this for 9 dishes. In each dish, we added 15 stentors. Then, we fed each dish 50 µL of algae solution. We placed 3 dishes in each incubator.

### *Data Collection*

Over three weeks, we took data for each dish. We counted the number of stentors and fed 50µL of the algae solution to each dish after counting. We did this three times a week for a total of nine data collections on Tuesdays, Thursdays, and Saturdays.

### *Statistical Analysis*

We used an ANOVA test to compare the means for each count per temperature across three groups. This test determines whether or not there is a statistically significant difference between the means of each of the three groups for which we collected data.

## Results

The purpose of this experiment is to examine the effects of temperature on the division process in *Stentor coeruleus.* The 20°C and 30°C groups showed a steep increase in the number of stentors present in the first ten days, nearly reaching 100 on average, but there was a sharp drop to around 30 stentors by day 20 (Figure 1, green line and orange line). The 10°C group increased slightly, to around 25 stentors, and then remained stagnant for the rest of the time (Figure 1, blue line). We performed an ANOVA test using the means from each of the three groups for each temperature. We found a statistically significant difference between the 10°C group and the 20°C group as well as the 10°C group and the 30°C group (Figure 1, $p = 0.008643$). The growth between the three groups under 10°C conditions remained constant with little variation (Figure 2). However, the 20°C and 30°C groups showed some variation between groups over time (Figure 3 & Figure 4). Overall, the stentors kept at warmer temperatures divided at a higher rate than those kept at colder temperatures. After the first ten days, the 30°C group averaged a total of 2.3833 cell divisions, 20°C averaged 2.2833, and 10°C averaged 0.7778 (Figure 5). However, we observed that the 30°C temperature resulted in the melting of several stentors by the end of the experiment (Figure 6).

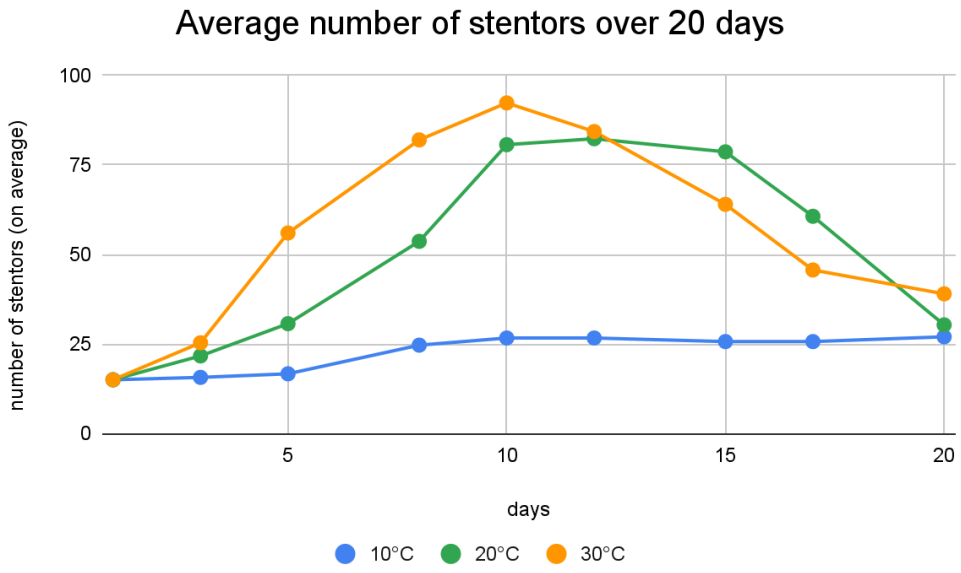


**Figure 1: Number of Stentors Present according to Temperature**

The blue line corresponds to the data collected for the 10°C group, the green is 20°C, and the orange is 30°C. Each line represents the average of three groups of data per temperature. The data is shown over twenty days with nine data points. The ANOVA test outputted a p-value of 0.008643, suggesting a statistically significant difference between the 10°C group and both the 20°C and 30°C, and no statistically significant difference was found between the 20°C and 30°C groups.

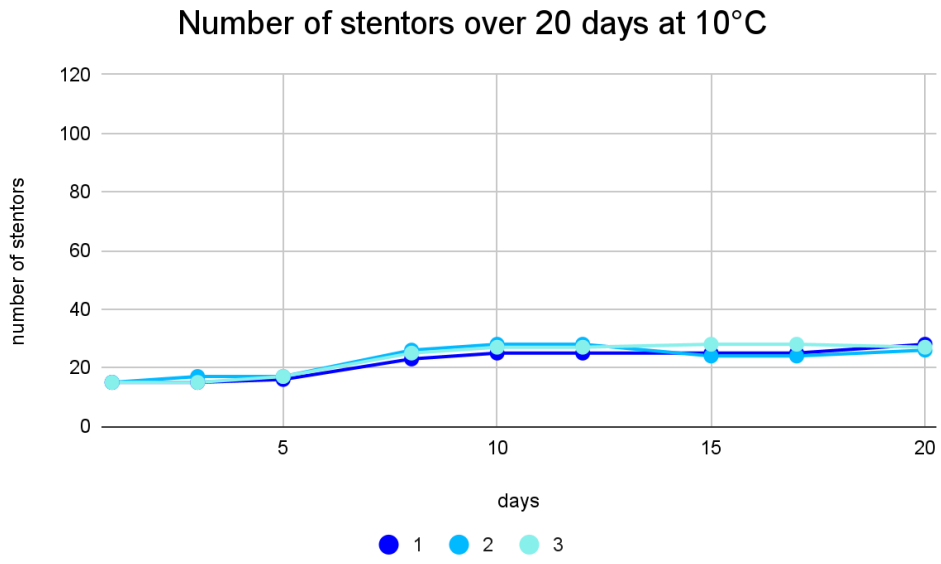

**Figure 2: Number of stentors at 10°C over three weeks**

The three lines correspond to the three replicate trials. The points represent the number of stentors in each dish over 20 days spent in an incubator at 10°C.

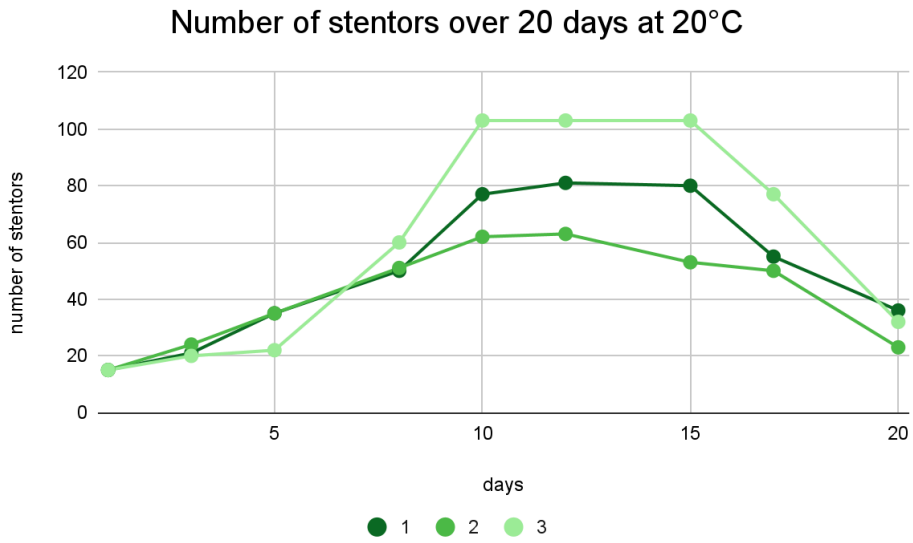


**Figure 3: Number of stentors at 20°C after three weeks**

The three lines correspond to the three replicate trials. The points represent the number of stentors in each dish over 20 days spent in an incubator at 20°C.

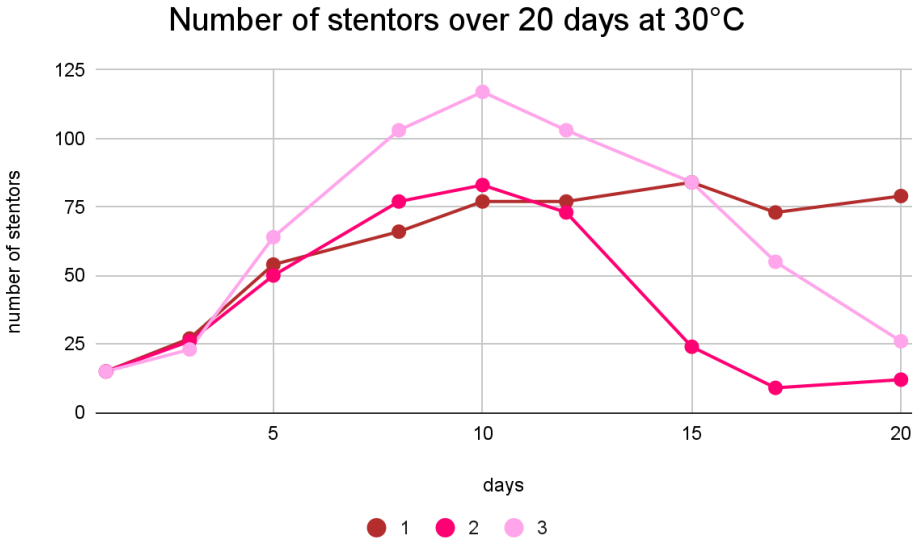


**Figure 4: Number of stentors at 30°C after three weeks**

The three lines correspond to the three replicate trials. The points represent the number of stentors in each dish over 20 days spent in an incubator at 30°C.

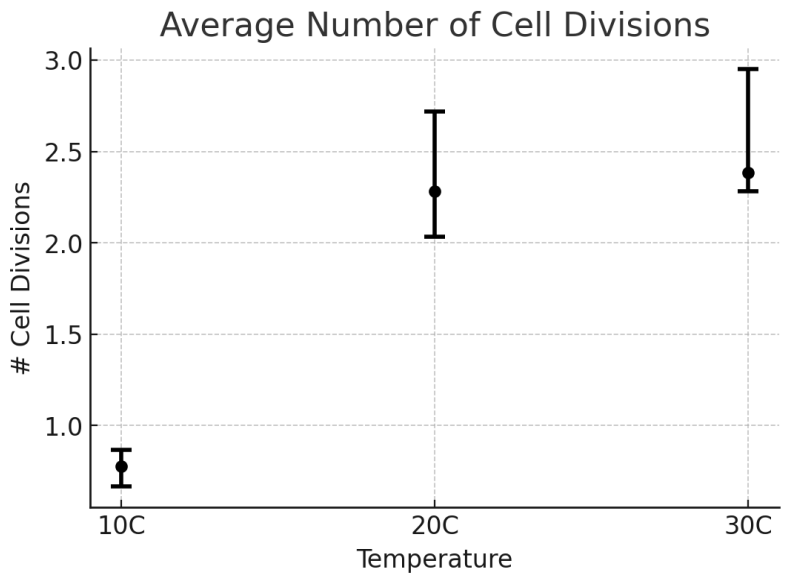


**Figure 5: Average number of cell divisions over 10 days**

The black dots represent the average cell division per temperature. The top error bar represents the most cell divisions, and the bottom error bar indicates the least cell divisions. Average cell divisions represented by dots: 10°C = 0.7778, 20°C = 2.2833, 30°C = 2.3833.

**Figure 6: Image of melted stentor**

This is an image taken through a microscope of a melted stentor after being exposed to 30°C for an extended period.

**Figure 7: Image of stentor in 30°C**

This is an image of a stentor taken through a microscope. We observed dark spots that may indicate the consumption of smaller cells by larger stentors.

**Discussion**

This experiment aimed to look at the effects of temperature on the division rates of *Stentor coeruleus* by comparing cell reproduction at 10°C, 20°C, and 30°C. Based on previous observations regarding temperature's impact on stentor movement and regeneration, it was hypothesized that lower temperatures would reduce division rates.

During the initial ten days, stentors at 20°C and 30°C exhibited average division rates of approximately 2.2833 and 2.3833 times, respectively, aligning with typical division frequencies of one to two times per week under standard conditions (approximately 25°C)(Figure 5). In contrast, stentors at 10°C only divided about 0.7778 times on average during the same period (Figure 5). These findings suggest that lower temperatures within the tested range decrease cell division in *S. coeruleus*. However, after ten days, a notable decline in stentor populations was observed at both 20°C and 30°C, with numbers decreasing to approximately 30 individuals by day 20. This unexpected reduction may be attributed to several factors. One possible explanation is temperature-induced stress. Prolonged exposure to elevated temperatures could have induced stress, impairing cellular functions and heat resistance. Previous studies have shown that temperature variations can significantly affect the metabolic activities and survival of ciliates,

including *Stentor coeruleus* (Laybourn, 1975). Another possible explanation is density-dependent inhibition. This phenomenon, where rapid population growth leads cells to compete for limited resources including nutrients, oxygen, and space, has been observed in various unicellular organisms (Billard, et al., 2020). One last potential explanation for the decline in stentors is that some single-celled organisms may exhibit cannibalistic behaviors in resource-scarce environments, although that is not always the cause (Tartar, 1961). While direct evidence of cannibalism in our stentors under these experimental conditions is lacking, the observed dark spots within larger stentors may indicate ingestion of smaller stentors (Figure 7). Further investigation is necessary to confirm this behavior. To further understand the relationship between temperature and stentor division rates, future experiments could optimize feeding regimens, explore intermediate temperatures, and investigate long-term effects.

Understanding how temperature influences cellular processes in *S. coeruleus* not only provides insights into the species' biology but also contributes to broader ecological knowledge, particularly in the context of climate change and its impact on aquatic microorganisms.

**References**

Billard, Bénédicte, et al. “Genetics and evolution of developmental plasticity in the nematode C. elegans: Environmental induction of the dauer stage.” *Biologie Aujourd'hui*, vol. 214, no. 1-2, 2020, pp. 45-53. *PubMed*, https://research.ebsco.com/c/oyqk56/viewer/pdf/jlwmu24oy5. Accessed 07 03 2025.

Laybourn, Johanna. “Respiratory Energy Losses in Stentor coeruleus Ehrenberg (Ciliophora).” *Oecologia*, vol. 21, no. 3, 1975, pp. 273-278. *Springer*, https://www.jstor.org/stable/4215187. Accessed 07 03 2025.

Schulte, Patricia M. “The effects of temperature on aerobic metabolism: towards a mechanistic understanding of the responses of ectotherms to a changing environment.” *Journal of Experimental Biology*, vol. 218, no. 12, 2015, pp. 1856-1866. *The company of Biologists*, https://journals.biologists.com/jeb/article/218/12/1856/13737/The-effects-of-temperature-on-aerobic-metabolism. Accessed 07 03 2025.

Tartar, Vance. *The Biology of Stentor*. Pergamon Press, 1961. *Internet Archive*, https://archive.org/details/biologyofstentor00tart. Accessed 07 03 2025.